

ME4453 Industrial Project Management

Semester 7

Locally Value Added Parts Design and Manufacturing

Final Report

By

Group 03

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1 Background Study

Sri Lanka's automotive assembly plants, such as Hyundai, Hero, and Micro, currently use locally value-added components. These assembly plants use about 30% of vehicle components locally sourced, including mudguards, seats, parcel trays, and fender covers. These are manufactured by third-party companies such as SW Industries and Ideal Industries.

However, the demand for locally value-added components is growing driven by,

- Government regulations requiring at least 20%–30% local value addition in assembly
- Increasing number of car repairs and after-market part replacements
- Demand for high-quality custom components in post-sale service centers

Despite this demand, limited availability, lack of standardization and separated manufacturing restrict the scalability of locally value-added components. This presents a high-potential business opportunity.

2 Project Introduction

The project “Locally Value Added Parts Design and Manufacturing” focuses on developing an integrated approach to design, manufacture, and supply locally produced automotive components for Sri Lankan vehicle assembly and service industries. The initiative is driven by the need to strengthen domestic manufacturing capabilities, minimize import dependency, and enhance quality consistency in locally fabricated vehicle parts.

This project aims to establish a structured business model that ensures efficient production and delivery of commonly required automotive parts such as mudguards, fender covers, and head restraints, using advanced manufacturing processes like industrial plastic injection molding. Through this approach, the project contributes to improving local value addition within the automotive sector and promoting sustainable industrial growth.

Beyond product development, the study evaluates the technical, social, ecological, and financial feasibility of initiating such a venture within Sri Lanka’s current industrial landscape. Key aspects include process optimization, cost analysis, quality management, and alignment with recognized manufacturing standards such as ISO 9001 and TS 16949.

Ultimately, this project seeks to demonstrate that with proper coordination, design innovation, and quality assurance, locally manufactured automotive parts can achieve competitive performance and market viability, supporting the long term vision of a self reliant automotive industry in Sri Lanka.

2.1 Project Scope & Target Audience

This project will initially focus on the design and manufacturing of locally value-added automotive components such as mudguards, wheel fender covers, seat components, headrests, parcel trays, and luggage trays, etc. The production process will involve the use of computer-aided design (CAD) tools such as SolidWorks, as well as mold-making, plastic injection molding, and sheet metal forming techniques. The main target audience includes local vehicle assembly plants such as Hyundai, Hero, and Micro, which require consistent supply of quality parts for their operations. In addition, the business will also serve local automotive aftermarket service centers and spare part retailers and garages across Sri Lanka, meeting the growing demand for repair and replacement parts in the local vehicle market.

2.2 Challenges

- Production managed by small, independent suppliers with limited coordination.
- Leads to inconsistent quality and reliability of components.
- Existing suppliers cannot expand capacity to meet the rising market demand.
- Despite local capability, many components are still sourced from abroad.

2.3 Aims and Objectives

2.3.1 Aim

To establish a local company that designs, manufactures, and supplies locally value-added components to Sri Lankan car assembly and service plants.

2.3.2 Objectives

- To design and manufacture commonly required LVA components
- To use fewer imported parts and make the products cheaper to produce
- To collaborate with local assembly plants
- To ensure consistent quality through ISO 9001/TS 16949 processes.

3 Conducting a Feasibility Study

3.1 Market Feasibility

3.1.1 Objectives of Market Feasibility

The primary objective of the Market Feasibility study was to validate that the projected demand for locally manufactured components is sufficient and sustainable, thereby justifying the required investment. This involved assessing the market size and growth, identifying and segmenting the target clientele, specifically the B2B assembly plants (Hyundai, Hero, Micro) and the robust aftermarket sector and analyzing the competitive landscape to pinpoint strategic gaps, particularly in quality and standardization. Ultimately, the study confirms that favorable regulatory policies, such as the Domestic Value Addition (DVA) mandate, create a high-potential environment for TCD Industries to secure long-term commercial relationships and eventually expand into regional export markets.

3.1.2 Market Opportunity and Regulatory Drivers

The market is poised for growth by enabling the replacement of imported components with locally manufactured high-quality alternatives. Government regulations currently mandate a minimum of 20% local value addition for four-wheel vehicles assembled under the Ministry of Industries' Standard Operating Procedure (SOP) to qualify for tax concessions. The SOP requires a detailed inspection process to verify this value, including a review of the Bill of Quantities (BOQ) and the component's manufacturing process. The government encourages local sourcing to achieve this DVA threshold, offering excise duty exemptions for vehicles with 30% domestic value addition. This policy has proven effective, with some local assembly projects, like MicroCars, achieving as much as 40% local cost content by manufacturing parts domestically. The overall auto component manufacturing sector is rapidly expanding, comprising approximately 50 companies. The primary opportunity for your project is to solve the current market issues of limited availability, lack of standardization, and separated manufacturing.

3.1.3 Target Audience (Customers)

Your product line (e.g., mudguards, seat components, parcel trays) targets two distinct customer bases.

- **Local Vehicle Assembly Plants (B2B):** The primary audience includes assembly operations such as Hyundai, Hero, and Micro [Project Brief]. These assemblers, along with over 15 other operating plants, require a continuous, reliable supply of standardized components to meet DVA requirements. They offer a steady B2B market for parts like seat components, bumpers, and parcel trays. [<https://investsrilanka.com/wp-content/uploads/2023/08/Auto-Components-Sector-2023.pdf>]
- **Local Automotive Aftermarket:** This substantial secondary market consists of local service centers, spare part retailers, and garages. This market segment is strong due to the large fleet of older and reconditioned imported vehicles on the roads driving up demand for maintenance and replacement parts. Customers often rely on professionals for repair and maintenance.

3.1.4 Competitive Strategy and Quality Advantage

Your competitive strategy centers on delivering internationally compliant quality and managing costs effectively.

- **Quality & Standardization (USP):** The core objective is to ensure consistent quality through ISO 9001/TS 16949 processes. The ISO/TS 16949 quality standard is a global industry prerequisite, often making certification essential for supplying OEMs. Adopting this system significantly reduces the risk of defects and variation in the supply chain, which is a major selling point to assemblers.
- **Cost Efficiency (The Value):** The objective is to use fewer imported parts to make the components cheaper to produce, thereby increasing local market competitiveness.
- **Competition:** While the industry has approximately 50 automotive component manufacturing companies, the specialization in standardized, high-quality plastic/metal parts targets the gap left by non-standardized local production.

3.1.5 Scalability and Export Potential

The business model is designed to support growth both regionally and globally.

- **Regional Exports:** The long-term plan is to explore future exports to other CKD-based plants in South Asia. This strategy is highly viable, as Sri Lankan firms benefit from

trade agreements that provide 100% duty-free access for many auto components to India and reduced duties to Pakistan under respective Free Trade Agreements.

- **Global Access:** The sector already has global recognition, with local firms exporting components to major markets, including the USA, Germany, and Japan.

3.2 Social Feasibility

3.2.1 Social Impact & Benefits

The establishment of the local manufacturing company is designed to have a strong positive social impact:

Job Creation: The operation of a local company aims to directly create jobs in the design, manufacturing, and logistics fields [Project Brief]. The broader Sri Lankan automotive sector currently supports over 18,000 workers, providing a foundation for growth in skilled employment.

Skill Development & Empowerment: The project will increase the demand for **technicians and skilled labor** in areas like computer-aided design (CAD), mold-making, plastic injection molding, and sheet metal forming [Project Brief]. This fosters the development of advanced industrial skills within the local community, contributing to human capital development.

National Economic Contribution: By establishing a local manufacturing base that targets 20%–30% local value addition and exploring exports, the project aligns with national goals to conserve foreign exchange and promote self-reliance in the industrial sector.

3.2.2 Stakeholder Engagement & Labor Standards

Successfully navigating the social landscape requires addressing labor and community relations:

- **Stakeholder Collaboration:** The business model centers on **collaboration with local assembly plants** such as Hyundai, Hero, and Micro for B2B supply, securing a long-term commercial relationship that benefits local partners.
- **Quality Assurance & Training:** The commitment to ensuring consistent quality through **ISO 9001/TS 16949 processes** implies an investment in rigorous training and transparent labor management necessary to maintain these international standards.

These standards require a systematic approach to quality management, which indirectly elevates local technical practices.

3.2.3 Potential Social Challenges

While the project is largely beneficial, potential challenges generally revolve around operational impact:

- **Land Use:** The project requires a small-scale factory setup, which necessitates appropriate land acquisition and zoning adherence to avoid conflict with residential or agricultural land use.
- **Environmental Perception:** Manufacturing processes, especially plastic injection molding and sheet metal forming, can face public scrutiny regarding waste and pollution; therefore, transparent compliance with local environmental regulations is critical to maintain positive community relations.

3.3 Technical Feasibility

3.3.1 Objectives of a Technical Feasibility

Concise and Direct (Recommended) Conducting the Technical Feasibility study is essential for defining the technical and operational blueprint of the manufacturing facility before the procurement stage. The core objective is to assess the technical requirements for designing and commissioning the Plastic Injection Molding and Sheet Metal Forming production lines. This analysis includes determining the precise specifications for machinery, evaluating raw material compatibility, and confirming that the integrated CAD-to-CAM workflow can achieve the required dimensional accuracy and consistent quality necessary for the automotive supply chain.

3.3.2 Plastic Component Manufacturing (Injection Molding)

The production of components like mudguards, wheel fender covers, and trays require plastic injection molding.

Table 1: Technical feasibility study for plastic component manufacturing

Parameter	Technical Details & Calculation
Material Selection	Primary material: ABS or Polypropylene (PP). These materials are preferred in automotive exterior and interior applications for their low weight, good impact resistance, and design flexibility.
Machinery Requirement (Clamping Force)	The primary technical metric is clamping force, measured in tons, which keeps the mold closed during injection. The required tonnage is typically calculated based on the part's projected area and the material properties. For general plastics like PP, a force of tons per square inch is a common estimate. Industrial machines suitable for automotive components usually require clamping forces of or more.
Material Cost (Estimation)	The current average market price for automotive-grade resins (ABS or PP) is approximately EUR 0.52 to 0.62 per kg (equivalent to Rs. 165 - Rs. 200 per kg at a conversion rate of 1 EUR = 320 LKR). This volatile raw material cost must be monitored closely to maintain the "cheaper to produce" objective.
Production Rate	Production rates are high, as injection molding cycles for mid-sized parts are generally fast, allowing for cost-effective, high-volume production. A cycle time of several seconds can yield 720 parts per hour for simple pieces.

3.3.3 Metal Component Manufacturing (Sheet Metal Forming)

The production of items like **seat frames** and support brackets requires **sheet metal forming** [Project Brief].

Table 2: Technical feasibility for metal component manufacturing

Parameter	Technical Details & Calculation
Machinery Requirement (Tonnage)	Production requires a (stamping press). The tonnage required is calculated based on the linear length of the cut, the material thickness, and the material's shearing strength. The general formula for blanking steel is: Tonnage $\approx$ Thickness $\times$ Linear Inches of Cut $\times$ 50,000. Industrial presses for automotive parts typically operate between \$100\text{ tons}\$ and \$500\text{ tons}\$.
Process Control	High-speed, high-volume production is achieved using Progressive Dies, which combine multiple cutting and bending operations into a single, automated process. The required dimensional accuracy (parallelism, perpendicularity) is critical for structural parts like seat frames.

3.3.4 Digital Design and Quality Control

The project hinges on a precise digital manufacturing workflow to ensure product quality and standardizationcan

- **CAD/CAM Integration:**

Computer-Aided Design (CAD) (e.g., SolidWorks) is crucial for precision modeling and creating dimensional specifications. Computer-Aided Manufacturing (CAM) software translates these models into machine instructions (G-code) for the CNC-controlled presses and mold-making equipment.

- **Design for Manufacturability (DFM):**

The technical team must perform DFM analysis during the design phase to identify and eliminate potential defects before expensive mold/tool creation begins, maximizing precision and cost-efficiency.

3.4 Ecological Feasibility

3.4.1 Objectives of Ecological Feasibility

The primary objective of the Ecological Feasibility study is to assess the project's environmental footprint and commitment to sustainable industrialization. This analysis focuses on minimizing the consumption of non-renewable energy by implementing energy-efficient machinery and exploring renewable energy integration (e.g., solar power) to offset factory power demands. Crucially, the study evaluates the comprehensive waste management plan for manufacturing byproducts, including the reprocessing of plastic scrap and the recycling of metal waste, while also establishing protocols to mitigate any negative impact on the local ecosystem, such as noise pollution and soil contamination from process lubricants.

3.4.2 Factors considered under Ecological Feasibility

a) Energy Consumption and Carbon Footprint Reduction

The manufacturing process, particularly injection molding and metal forming, are energy-intensive

- **Energy Optimization:** Implementing energy-efficient machinery (e.g., servo-driven injection molding machines, which can reduce energy consumption by 20% to 80% compared to conventional hydraulic machines) is critical. This approach reduces the reliance on fossil-fuel-based electricity and the overall carbon footprint.
- **Renewable Energy Integration:** Exploring the feasibility of integrating solar power into the facility's energy mix can further reduce greenhouse gas emissions and operational costs, contributing to cleaner production.

b) Waste Management and Pollution Prevention

Effective management of waste streams is crucial to prevent environmental contamination.

- **Plastic Waste Management:**
 - **Recycling of Plastic Scrap:** The project should implement a robust system for reprocessing plastic runners and rejected parts (e.g., ABS, PP) back into the manufacturing cycle. This significantly reduces plastic waste and raw material costs.

- Minimizing Waste: Optimized mold designs (e.g., hot runner molds) can drastically reduce plastic scrap (runners) by up to 20% - 50%.
- **Metal Waste Management:**
 - Scrap Metal Recycling: Implementing processes for collecting and recycling metal scraps (steel, aluminum) from the stamping process. The global steel recycling rate is high (around 85% for automotive steel), underscoring its environmental benefit.
- **Hazardous Waste:** Proper handling and disposal of any lubricants, coolants, or chemical by-products from metal treatment processes are necessary to prevent soil and water contamination.

c) Resource Consumption (Water & Raw Materials)

Sustainable consumption of resources is vital for long-term ecological viability.

- **Water Management:** Industrial manufacturing typically requires water for cooling, cleaning, and sometimes material processing. Implementing closed-loop cooling systems for machinery can significantly reduce water consumption and discharge.
- **Sustainable Sourcing:** Prioritizing raw material suppliers (plastic resins, metal sheets) who adhere to environmental standards (e.g., ISO 14001 certified) supports a greener supply chain.

d) Impact on Local Ecosystems

The physical footprint and operational aspects of the facility require careful consideration to minimize local ecological disruption.

- **Land Use:** The construction of a small-to-medium scale manufacturing plant requires careful site selection to avoid ecologically sensitive areas, local flora, and fauna [Project Brief]. An Environmental Impact Assessment (EIA) may be necessary depending on the scale and location, as per local regulations.
- **Noise and Air Quality:** Manufacturing operations can generate noise and, potentially, air emissions (e.g., from paint booths if added later). Proper ventilation systems and

noise abatement measures are crucial to minimize impact on surrounding communities and ecosystems.

3.5 Financial Feasibility

4 Financial Feasibility Study

4.1 Objectives of Financial Feasibility

The main objective of this financial feasibility study is to determine the financial viability of establishing a locally value-added automotive component design and manufacturing facility in Sri Lanka.

- Evaluate the required initial investment for plant setup, equipment, and tooling.
- Estimate annual operating and maintenance costs.
- Project cash inflows and outflows over a five-year period.
- Assess financial indicators such as Net Present Value (NPV), Internal Rate of Return (IRR), and Payback Period.
- Recommend whether the proposed investment is financially feasible.

4.2 Cost Breakdown of Initial Investment

Table 3: Initial Investment

Cost Center	Cost Item / Components	Cost ('000) LKR
Factory Setup & Infrastructure	Lease deposit (6 months)	2,100
	Factory interior preparation (power, compressed air, lighting, ventilation)	1,000
	Electrical & safety installations (3-phase wiring, fire safety, signage)	600
	Office furniture, storage, canteen setup	300
	Utility connection deposits (electricity, water, internet)	200

Cost Center	Cost Item / Components	Cost ('000) LKR
Subtotal – Factory Setup & Infrastructure		4,200
Machinery & Production Equipment	150T injection molding machine	4,500
	80T press forming machine	2,000
	Foam molding & curing setup (for headrests)	1,500
	Compressors, chillers, power tools, hand tools	800
	Molds & dies (mudguards, fender covers, headrests)	2,500
Subtotal – Machinery & Equipment		11,300
Raw Materials & Initial Stock	Plastic granules (PP, ABS, PU)	1,000
	Steel rods/tubes (for headrest frames)	500
	Foam, adhesives, paint, finishing materials	400
	Consumables & chemicals	200
Subtotal – Raw Materials		2,100
CAD, QA & Testing Tools	SolidWorks license + workstation	1,500
	Measuring instruments (calipers, gauges, hardness tester)	250
	Load/strength test rigs	300
Subtotal – CAD & QA Tools		2,050

Cost Center	Cost Item / Components	Cost ('000) LKR
Labor & Salaries (Pre-production phase)	Project Manager	960
	Mechanical Design Engineer	720
	Production Supervisor	480
	Machine Operators (5 persons)	1,200
	QA & Maintenance Technicians (2 persons)	480
	Administrative & support staff	240
Subtotal – Labor & Wages		4,080
Packaging & Logistics Setup	Packaging tables, crates, labeling setup	250
	Initial transport & delivery arrangement	300
Subtotal – Packaging & Logistics		550
Registration, Legal & Marketing	Business registration, ISO documentation	120
	Environmental & EPL license	80
	Marketing, website, product brochures	200
Subtotal – Registration & Marketing		400
Contingency (10%)	Allowance for unforeseen expenses	2,270
Total Project Cost ('000) LKR		27,000

4.3 Cash Flow During Subsequent Years

When calculating the cash flow these assumptions were made.

- The local parts are manufactured for a vehicle lot on assembly plants for each month.
- The car lot is same for all 12 months of the year.
- The selling price is determined by the local parts for that vehicle lot size.

The main cash inflow is from the sales of local value-added parts mudguards, fender covers, and head restraints for local car assembly plants and repair centers.

Also, the economic useful life of the project was defined to be 10 years. The main reason is the automotive industry in Sri Lanka is developing in further years. Therefore, the demand on manufacturing local value added parts only can be a constraint for the expansion of the business. Therefore, 10 years is considered as the economic useful life.

ABANS SRILANKA						
418 ITEMS 37,920 PCS 257 CASES						
AI3 PE- 60NOS BRAND NEW CKD KITS FOR HYUNDAI GRAND 110 1.2L AUTOMATIC CIF COLOMBO SRI LANKA AS PER PROFORMA INVOICE NO.GBH/PI/AI3 PE/PO-03-LOT-08-LP - 60 UNITS DATED 30/07/2024						
Net Weight : 30326.29 KG Gross Weight : 43193.73 KG CBM : 564.894 MADE IN INDIA	DESCRIPTION	Carsets	FOB @USD	FREIGHT @ USD	INSURANCE @USD	TTL CIF @ USD
	AI3 PE- 60NOS BRAND NEW CKD KITS FOR HYUNDAI GRAND 110 1.2L AUTOMATIC	60 Units	307,274	7,500	94.20	314,868
DECLARATION : We declare that the invoice shows the actual price of the goods described and that all particulars are true & correct . Certified that the description/weight/quantity/and value of the goods covered by this invoice and packing list have been verified and found correct.The goods were stuffed into container nos and sealed with self seal no as below under my supervision. "as per details given below at the factory premises of M/s.Glovis India Pvt Ltd.						

Figure 1: Lot Size of assembly plants

Cash Inflows

The monthly production capacity is determined by the demand of the vehicle assembly plants and automotive repair centres. Also, the capacity per month is determined by the lot size, which means the number of vehicles each will be released for the market. Therefore, the capacity was obtained as follows.

Table 4: Production capacity

Production Capacity (Per month)	Qty
Hyundai	50
Micro	50
Hero	50

Mahindra	50
United Motors	50
Repair centres	400

The main product that our business will overgo is,

- Mudguards
- Fender covers
- Head restraints

Table 5: Annual Revenue

Item	Qty	Selling Price(LKR)/ LOT	Annual Revenue(LKR)
Mudguards	2600	500,000.00	6000000.00
Fender covers	2600	500,000.00	6000000.00
Headrestraints	3250	700,000.00	8400000.00
Total Revenue / Year			20,400,000.00

$$\text{Number of cars/month} = 50$$

$$\text{Mudguards per car} = 4$$

$$\text{Mudguards per month} = 4 * 50 = 200$$

$$\text{Mudguards per year} = 200 * 12 = 2400$$

$$\text{For repair centres} = 200$$

$$\text{Total mudguards/ month} = 2600$$

The selling price is obtained for the annual quantity of mudguards. Therefore,

$$\text{Annual revenue/ mudgaards} = 2600 * 500,000 = \text{Rs. } 6,000,000.00$$

Similarly, the other cash inflows from other products are also obtained from the given table.

$$\text{The annual cash inflow} = \text{Rs. } 20,400,000.00$$

When obtaining the cash inflows for the next years the cash inflow will be considered a 5% increment. However, the cash inflow can be vary with demand variations due to different conditions.

Cash Outflows

Cash outflows include raw materials, labor, maintenance, logistics, and administration.

Table 6: Operating Cost

Raw materials & consumables	2,100,000.00
Labor & salaries	4,080,000.00
Utilities, maintenance & overheads	200,000.00
Marketing, admin & transport	400,000.00
	6,780,000.00

The cash outflow for CAD softwares such as SolidWorks can be neglected for the 1st year as SolidWorks provide the first year free for startups. However, relevant cost values will be charged for the next year. The first 3 year's subscription plan is shown below of SolidWorks.

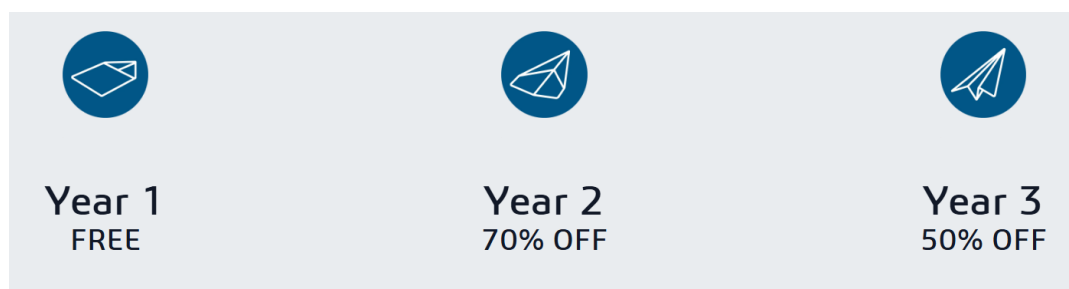


Figure 2: Solidworks subscription for start ups

The cash outflow will be assumed to be increased by 5% as well due to the variations that can be overcome on the next years.

Weighted Average Cost of Capital (WACC) or Discount rate (r)

The expectation on the methods of funding for the business is mainly focused on,

- Debt financing(D) – Money will be borrowed from potential funding bodies and will be repaid with the interest.

Ex-: Bank loans

- Equity financing(E) – Selling the business ownership shares of the company to possible investors and the profits will be shared accordingly.

The initial investment of the business will be mainly focused on these two financing methods. Therefore, 60% of the initial investment (Rs. 27,000,000) is debt financed while the rest of will be funded through equity financing.

Table 7: Investement types

Initial investement from Debt financing	16,200,000.00
Initial investement from Equity financing	10,800,000.00

As a startup business there are many funding bodies for SME businesses. As per the SME loans from people's bank the interest rates for,

- AWPLR + 2.5% p.a. Monthly Review (No Floor Rate) (Average interest rate)

Based on these values the weighted average cost of capital of the business can be calculated as below.

$$\text{Average cost of debt}(r_d) = 2.5\%$$

$$\text{Average cost of equity}(r_e) = 11\%$$

$$\text{Weighted average cost of capital} = \left(\frac{E}{E + D} * r_e \right) + \left(\frac{D}{E + D} * r_d \right)$$

$$\text{Weighted average cost of capital} = \left(\frac{40}{40 + 60} * 11 \right) + \left(\frac{60}{40 + 60} * 2.5 \right)$$

$$\text{Weighted average cost of capital} = 7.6 \%$$

Calculation of Depreciation

Depreciation is calculated on these,

- Machinery and equipment used in manufacturing
- Molds, dies, jigs and tools
- Quality assurance instruments (Ex:- Vernier caliper)
- Factory buildings

The annual depreciation is calculated for the above cost elements as below. The depreciation rates are obtained from the Inland Revenue Department of Sri Lanka.

Table 8: Depreciation Items

Depreciation Item	Cost	Depreciation Rate	Annual depreciation
Machinery and equipment used in manufacturing	11,300,000.00	0.333	3,762,900.00
Molds, dies, jigs and tools			
Computers, CAD software & QA instruments	550,000.00	0.25	137,500.00
Furniture and office equipment	4,200,000.00	0.1	420,000.00
Factory buildings or permanent improvements			

- Machinery and equipment used in manufacturing – 33%
- Molds, dies, jigs and tools – 33%
- Quality assurance instruments (Ex:- Vernier caliper) – 25%
- Factory buildings – 10%

The annual depreciation is calculated using the reducing-balance method.

$$NBV = \text{Cost} \times (1 + \text{Depreciation rate})^n$$

NBV – Net book Value

Cost – Original asset cost

Depreciation rate – Reducing balance depreciation rate per period

n – Number of periods

Sample calculation for Machinery and equipment used in manufacturing,

$$NBV = 11,300,000 \times (1 + 33\%)^{10} = \text{Rs. } 3,762,900$$

4.4 Net Cashflow

Cash Inflow for the 1st year = Rs. 20,400,000.00

Cash outflow for the 1st year = Operating cost + Depreciation cost

$$\text{Cash outflow} = 6,780,000.00 + 11,729,000.00$$

$$\text{Cash outflow} = 18,509,600.00$$

Assume the cash inflow and cash outflow increase annually by 5%.

Table 9: Net Cash flow

Year	Revenue (LKR)	Operating Cost (LKR)	Depreciation (LKR)	Net Income (LKR)	Net Cash Flow (LKR)
0	-	-	-	-	- 27,000,000.00
1	20,400,000.00	6,780,000.00	11,729,600.00	1,890,400.00	1,890,400.00
2	21,420,000.00	7,119,000.00	8,738,620.70	5,562,379.30	5,562,379.30
3	22,491,000.00	7,474,950.00	6,647,004.13	8,369,045.87	8,369,045.87
4	23,615,550.00	7,848,697.50	5,166,209.75	10,600,642.75	10,600,642.75
5	24,796,327.50	8,241,132.38	4,102,365.31	12,452,829.82	12,452,829.82
6	26,036,143.88	8,653,188.99	3,324,964.13	14,057,990.75	14,057,990.75
7	27,337,951.07	9,085,848.44	2,745,943.96	15,506,158.67	15,506,158.67
8	28,704,848.62	9,540,140.87	2,305,699.51	16,859,008.25	16,859,008.25
9	30,140,091.05	10,017,147.91	1,963,726.94	18,159,216.21	18,159,216.21
10	31,647,095.61	10,518,005.30	1,692,363.17	19,436,727.13	19,436,727.13

4.5 Simple Payback Period

Table 10: Simple Payback Period

Year	Period	Net Cash flows (LKR)	Cumulative Cash flows (LKR)
2025	0	-27,000,000.00	-27,000,000.00
2026	1	1,890,400.00	-25,109,600.00
2027	2	5,562,379.30	-19,547,220.70
2028	3	8,369,045.87	-11,178,174.83
2029	4	10,600,642.75	-577,532.08
2030	5	12,452,829.82	11,875,297.73
2031	6	14,057,990.75	25,933,288.48
2032	7	15,506,158.67	41,439,447.15
2033	8	16,859,008.25	58,298,455.40
2034	9	18,159,216.21	76,457,671.61
2035	10	19,436,727.13	95,894,398.73

The simple payback period shows how many years and months it takes to cover the initial investment of the project.

Therefore, the payback period on the business = **4 years and 6months**

4.6 Net Present Value (NPV)

Table 11: Net Present Value

Year	Period	Net Cash flows (Future value of cash inflows)	Discount factor (r - 7.4%)	Present Value (PV)
2025	0	-27,000,000.00	1	-27,000,000.00
2026	1	1,890,400.00	0.931098696	1,760,148.98
2027	2	5,562,379.30	0.866944783	4,822,275.71
2028	3	8,369,045.87	0.807211157	6,755,587.20
2029	4	10,600,642.75	0.751593256	7,967,371.60
2030	5	12,452,829.82	0.699807501	8,714,583.71
2031	6	14,057,990.75	0.651589852	9,160,044.11
2032	7	15,506,158.67	0.606694462	9,407,500.59
2033	8	16,859,008.25	0.564892422	9,523,526.01
2034	9	18,159,216.21	0.525970598	9,551,213.81
2035	10	19,436,727.13	0.489730538	9,518,758.84
Total PV of inflows				50,181,010.56
Total PV of inflows		50,181,010.56		
Initial investment (I ₀)		27,000,000.00		
NPV(Net Present Value)		23,181,010.56		

The NPV is calculated using the formula.

$$PV = DF \times FV$$

$$\text{Discounting Factor}(DF) = \frac{1}{(1+r)^n}$$

$$NPV = \sum_{i=0}^n \frac{FV_i}{(1+r)^i} - I_0$$

PV - Present Value

FV - Future Value of cash inflows

I₀ - Initial investment or cash outlay

r - Discount Rate (Usually taken the firm's average cost of capital, represents the risk involved in the investment)

n - Number of periods (usually years) in which the future cash flows will occur

The NPV value is positive. Therefore, the business is financially achievable.

4.7 Discounted Payback Period

Table 12: Discounted payback period

Year	Period	Net Cash flows (Future value of cash inflows)	Discount factor (r - 7.4%)	Present Value (PV)	Cumulative PV
2025	0	-27,000,000.00	1.00	-27,000,000.00	-27,000,000.00
2026	1	1,890,400.00	0.93	1,760,148.98	-25,239,851.02
2027	2	5,562,379.30	0.87	4,231,886.32	-21,007,964.71
2028	3	8,369,045.87	0.81	5,628,680.06	-15,379,284.65
2029	4	10,600,642.75	0.75	6,353,811.74	-9,025,472.91
2030	5	12,452,829.82	0.70	6,660,512.15	-2,364,960.76
2031	6	14,057,990.75	0.65	6,708,140.90	4,343,180.14
2032	7	15,506,158.67	0.61	6,597,229.58	10,940,409.71
2033	8	16,859,008.25	0.56	6,391,362.61	17,331,772.33

2034	9	18,159,216.21	0.53	6,130,856.92	23,462,629.24
2035	10	19,436,727.13	0.49	5,841,328.00	29,303,957.25

The discounted payback period = **5years and 4months**

4.8 Internal Rate of Return (IRR)

Table 13: Internal Rate of Return

Year	Period	(Future value of cash inflows)	Discount factor (r - 7.4%)	Present Value (PV)	Discount factor (r - 30%)	Present Value (PV)
2025	0	-27,000,000.00	1.00	-27,000,000.00	1	-27,000,000.00
2026	1	1,890,400.00	0.93	1,760,148.98	0.769230769	1,454,153.85
2027	2	5,562,379.30	0.87	4,231,886.32	0.591715976	3,291,348.70
2028	3	8,369,045.87	0.81	5,628,680.06	0.455166136	3,809,306.27
2029	4	10,600,642.75	0.75	6,353,811.74	0.350127797	3,711,579.69
2030	5	12,452,829.82	0.70	6,660,512.15	0.269329074	3,353,909.13
2031	6	14,057,990.75	0.65	6,708,140.90	0.207176211	2,912,481.26
2032	7	15,506,158.67	0.61	6,597,229.58	0.159366316	2,471,159.38
2033	8	16,859,008.25	0.56	6,391,362.61	0.122589474	2,066,736.95
2034	9	18,159,216.21	0.53	6,130,856.92	0.094299595	1,712,406.74
2035	10	19,436,727.13	0.49	5,841,328.00	0.07253815	1,409,904.23
Net Present Value				29,303,957.25		-807,013.80

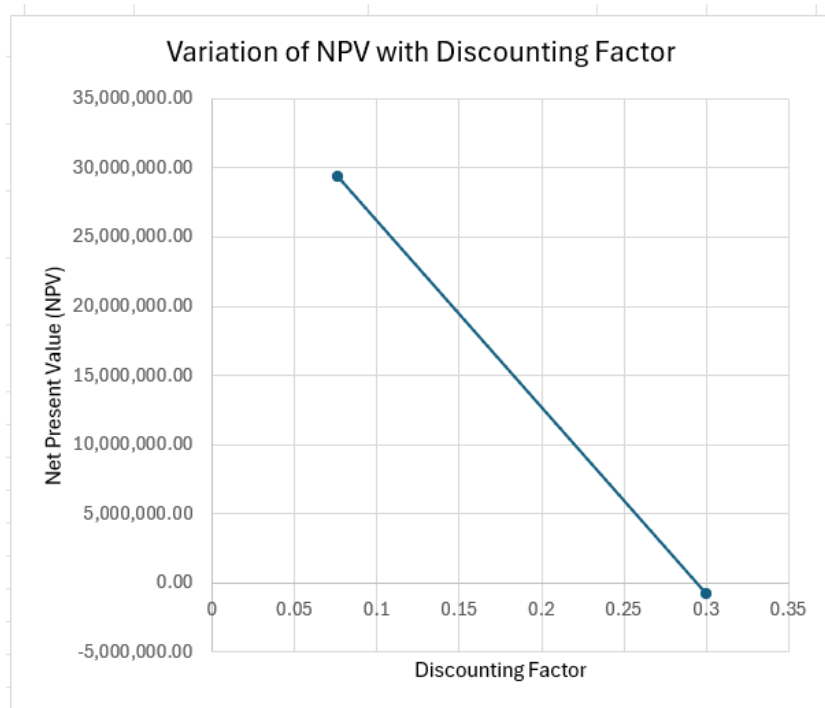


Figure 3: NPV vs Discounting factor

X (Discounting factor)	Y (Net present Value)
0.076	29,303,957.25
0.3	-807,013.80

Internal Rate of Return (IRR)=13%

Cost of capital = 7.6%

Since $IRR > \text{Cost of capital}$

The business is feasible and possible for acceptance

5 Implementation Plan

5.1 Methodology

The TCD Industries project implementation utilized a systematic decomposition approach, comprising six principal tasks. These tasks were further divided into sub-tasks, ensuring optimal resource allocation and accurate forecasting of project milestones. The key task areas are listed below.

1. Project Initiation
2. Feasibility study and Site selection
3. Designing and Planning
4. Funding and Partnerships
5. Procurement and Tendering
6. Construction and Installation
7. Community Training
8. Project Closure

TASK NAME	DURATION
Phase 1: Project Initiation & Feasibility (50 Days)	40 days
1.1 Define Project Scope, Aim, and Deliverables	5 days
1.2 Conduct Feasibility Study (Market, Technical, Social, Ecologi...	15 days
1.3 Identify and Secure Project Team Roles (Resources)	10 days
1.4.1 Develop Detailed Cost Breakdown & Cash Flow Projections	5 days
1.4.2 Prepare Funding Proposal	5 days
1.4.3 Secure Investment/Loan Approval	10 days
Phase 2: Detailed Design and Tooling Prep (40 Days)	45 days
2.1 Select Final LVA Components and Detailed Specs	10 days
2.2.1 Initial Part Design - Plastic Components (CAD)	8 days
2.2.2 Initial Part Design - Metal Components (CAD)	7 days
2.3 Design Review and Material Selection Finalization	5 days
2.4 Finalize Designs and Mold/Tooling Specifications	15 days
Phase 3: Procurement and Factory Setup (75 Days)	65 days
3.1 Finalize Contracts with Raw Material Suppliers (Steel/Plastic)	10 days
3.2 Procure and Order Heavy Machinery (Press, Injection Moldi...	10 days
3.3.1 Site Identification and Acquisition/Lease	16 days
3.3.2 Civil Works & Utility Setup (Power, Water)	20 days
3.4.1 Mold Design Final Review & Sign-off	5 days
3.4.2 Mold Fabrication Process	35 days
3.5 Delivery and Installation of Machinery	10 days

Phase 4: Commissioning and Quality Assurance (60 Days)	64 days
4.1 Initial Machine Commissioning and Calibration	5 days
4.2 First Component Trial Run	10 days
4.3.1 Material Verification and Load/Fit Testing	10 days
4.3.2 Dimensional Inspection & Tooling Adjustment/Revisions	10 days
4.4 Second Component Trial Run	10 days
4.5 Finalize Production Protocols and QA Documentation (ISO A...	15 days
Phase 5: Production & Market Integration (35 Days)	35 days
5.1 Train Production Staff and Operators	10 days
5.2.1 Develop Marketing Material/Samples for Assemblers	10 days
5.2.2 B2B Negotiation & Final Contract Signing	15 days
5.3 First Batch Delivery to Assembly Plants	10 days
Phase 6: Project Closure & Scale-up (20 Days)	16 days
6.1 Collect Client Feedback and Conduct Post-Project Review	10 days
6.2.1 Document 'Lessons Learned' and Future Recommendations	6 days

Figure 17: List of Tasks and their durations

5.2 Work Break Down Structure (WBS)

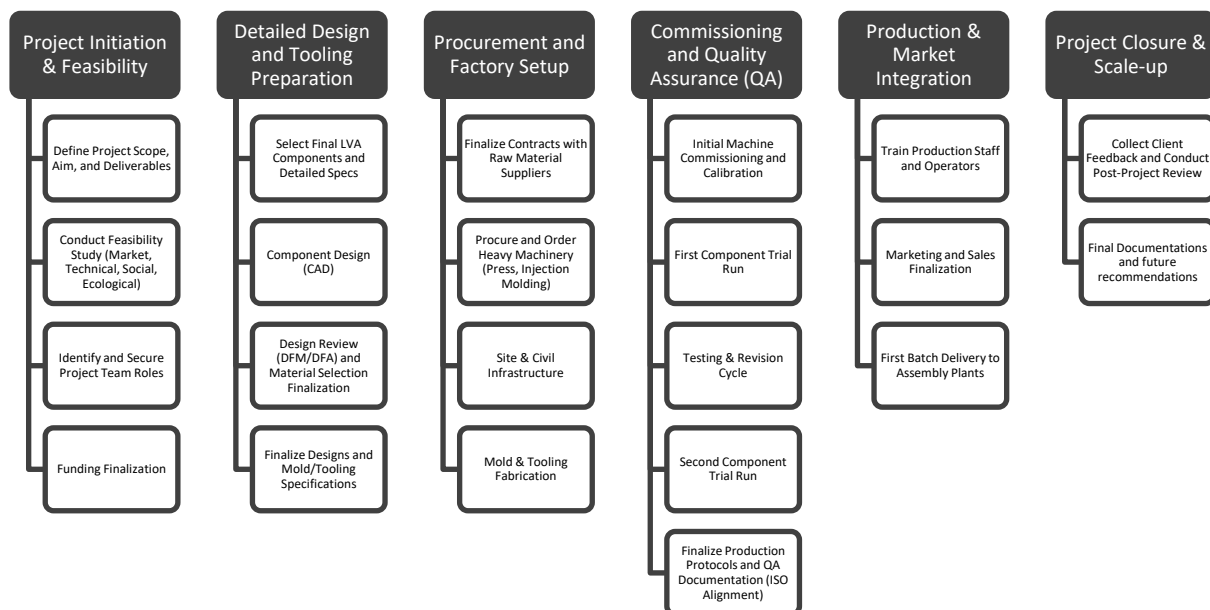


Figure 18: Work Breakdown Structure

6 Potential Funding Bodies

Securing the required capital investment of LKR \$27 Million is vital for project launch. Given that TCD Industries is a commercial, technology-driven manufacturing enterprise, potential financing is categorized across three pillars: commercial lending, government-backed development schemes, and institutional support. The following national and international organizations offer viable credit lines and support programs where the Ministry of Industries or the Industrial Development Board (IDB) can act as an intermediary, facilitating due diligence and ensuring compliance with national economic objectives.

Table 14: Potential Funding Bodies

Type of Funding Body	Specific Schemes / Programs
Government-Backed Development Loans	SMILE III (Small & Micro Industries Leader & Entrepreneur Promotion) / E-Friends II (Environmentally Friendly Solution Fund) Loan Schemes - https://www.peoplesbank.lk/sme-loans/
National Commercial Banks	People's Bank, Bank of Ceylon (BOC), DFCC Bank, HNB, Commercial Bank, SDB Bank
Development Banks (International)	Asian Development Bank (ADB) & World Bank Credit Lines (via local PFIs)
Industrial Support Institutions	Industrial Development Board (IDB) / Ministry of Industries
Export Promotion	Sri Lanka Export Development Board (EDB)

7 Risk associated with the project

Technical & Quality Control

Tooling Failure and Part Variation: Technical failure of custom injection molds or press dies (the equivalent of the solar components) resulting in components that do not meet the precise ISO/OEM dimensional specifications.

Process Inconsistency: Difficulty in maintaining the required quality standards (ISO 9001/TS 16949) across high-volume production cycles, leading to high scrap rates of plastic or metal parts and potential rejection by B2B clients.

Supply Chain & Operational

Raw Material Volatility: Unexpected, sharp price increases in imported raw materials like polymer resins (ABS/PP) or steel sheets, which directly erodes the project's core competitive advantage of being "cheaper to produce" than imports.

Machinery Downtime: Operational or technical failure of key equipment—specifically the Industrial Injection Molder or Hydraulic Press—due to a lack of specialized local maintenance expertise or critical spare parts.

Financial & Project Governance

Contingency Exhaustion: Extra expenses due to unforeseen events (cost overruns in mold development or factory setup) deplete the 10% contingency budget, ultimately requiring re-financing or delaying project completion.

Commercial Contract Loss: Failure to secure or retain long-term supply agreements with local assembly plants, leading to unused production capacity and low cash flow.

$$\text{Risk} = \text{Likelihood} \times \text{Impact}$$

Table 15: Risk Assessment

Category	Potential Risk Event	Likelihood	Impact	Risk Level
Technical / Production	Equipment failure	Medium	High	High
	Incorrect CAD drawings	Medium	Medium	Moderate
Supply Chain & Material	Delay or shortage of raw materials	High	Medium	High
	Price variations in imported raw materials	Medium	High	High
Financial / Economic	Cost overruns during setup or operations	Medium	High	High
	Changes in exchange rates	Low	Medium	Moderate
	Demand drop in automotive industry	Low	High	Moderate
Human Resource / Skills	Operator errors or poor training	Medium	Medium	Moderate
	Staff turnover of key engineers	Low	Medium	Low
Environmental / Safety	Fire or flood damage in factory	Low	High	Moderate
	Air pollution or waste disposal non-compliance	Medium	Medium	Moderate
Market & Customer	Customer rejection due to quality defects	Medium	High	High
	Late deliveries for suppliers	Low	High	Moderate

8 Project Timeline

8.1 Gantt Chart

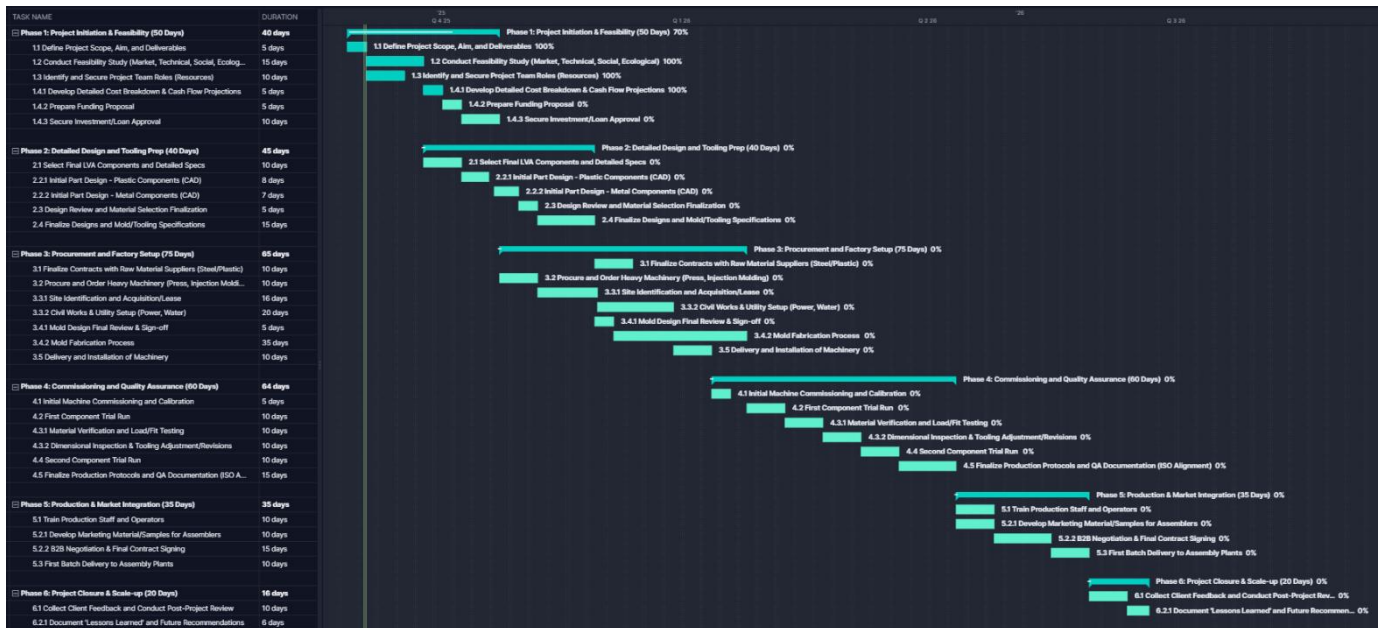


Figure 20: Gantt Chart

8.2 Network Diagram

The following table shows the task corresponding to each letter in the Network Diagram.

Table 16: Critical path method calculation

ID	Task Name	Duration (D)	Predecessor	EST	EFT	LST	LFT	Total Float (TF)
A1	Define Project Scope, Aim, and Deliverables	5	-	0	5	0	5	0
A2	Conduct Feasibility Study	20	A1	5	25	10	30	0
A3	Identify and Secure Project Team Roles	10	A1	5	15	5	15	0
A4.1	Develop Detailed Cost Breakdown & Cash Flow	5	A2	25	30	30	35	5
A4.2	Prepare Funding Proposal/Pitch Deck	5	A4.1	30	35	35	40	5
A4.3	Secure Investment/Loan Approval	10	A4.2	35	45	40	50	5
B1	Select Final LVA Components and Specs	15	A2	25	40	25	40	0
B2.1	Initial Part Design - Plastic Components (CAD)	8	B1	40	48	40	48	0
B2.2	Initial Part Design - Metal Components (CAD)	7	B2.1	48	55	48	55	0
B3	Design Review (DFM/DFA)	5	B2.2	55	60	55	60	0

B4	Finalize Designs and Mold/Tooling Specs	10	B3	60	70	60	70	0
C1	Finalize Contracts with Raw Material Suppliers	10	B4	70	80	70	80	0
C2	Procure and Order Heavy Machinery	10	A4.3	45	55	50	60	5
C3.1	Site Identification and Acquisition/Lease	15	C2	55	70	60	75	5
C3.2	Civil Works & Utility Setup	20	C3.1	70	90	75	95	5
C4.1	Mold Design Final Review & Sign-off	5	B4	70	75	70	75	0
C4.2	Mold Fabrication Process (Long Lead)	35	C4.1	75	110	75	110	0
C5	Delivery and Installation of Machinery	10	C3.2	90	100	95	105	5
D1	Initial Machine Commissioning and Calibration	5	C5	100	105	105	110	5
D2	First Component Trial Run (T1)	10	D1, C4.2	110	120	110	120	0
D3.1	Material Verification and Load/Fit Testing	10	D2	120	130	120	130	0
D3.2	Dimensional Inspection & Tooling Adjustment/Revisions	10	D3.1	130	140	130	140	0
D4	Second Component Trial Run (T2)	10	D3.2	140	150	140	150	0
D5	Finalize Production Protocols & QA Docs (ISO Alignment)	15	D4	150	165	150	165	0
E1	Train Production Staff and Operators	10	D5	165	175	180	190	15
E2.1	Develop Marketing Material/Samples	10	D5	165	175	165	175	0
E2.2	B2B Negotiation & Final Contract Signing	15	E2.1	175	190	175	190	0
E3	First Batch Delivery to Assembly Plants	10	E1, E2.2	190	200	190	200	0
F1	Collect Client Feedback and Post-Project Review	10	E3	200	210	200	210	0
F2.1	Document 'Lessons Learned' and Future Recommendations	5	F1	210	215	210	215	0
F2.2	Prepare Final ME4453 Module Report	5	F2.1	215	220	215	220	0

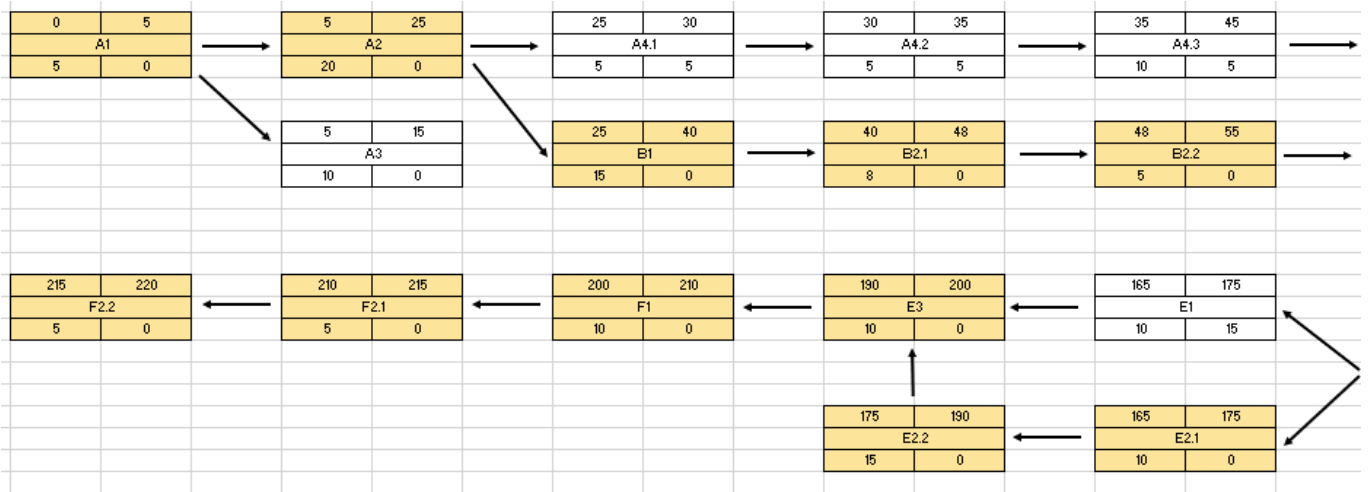
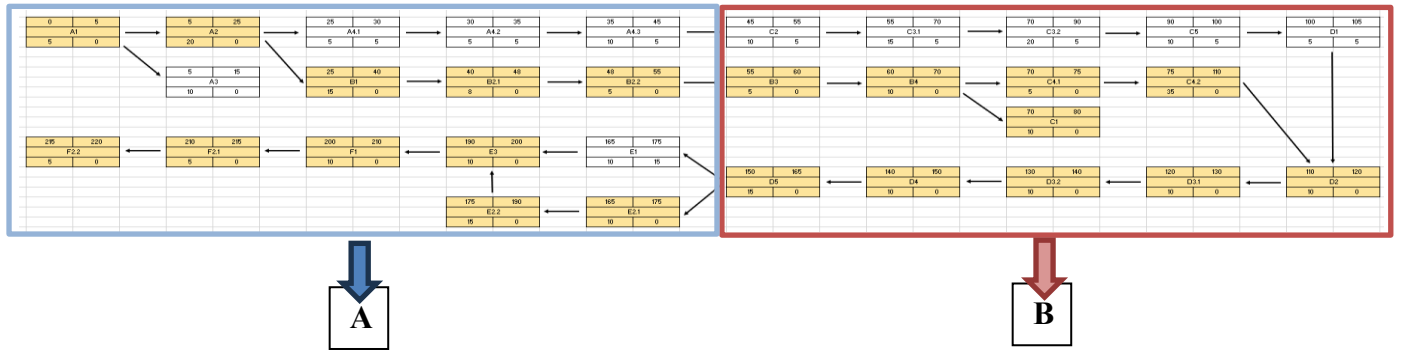


Figure 21: Enlarged View of A

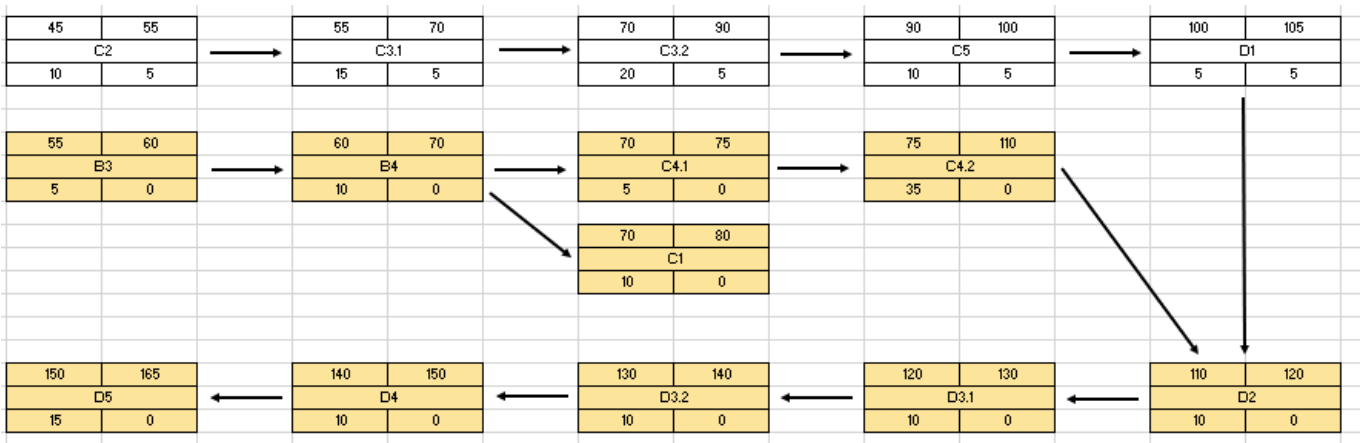


Figure 22: Enlarged View of B

Focused and Direct Based on the Network Diagram analysis, two critical paths were identified. The critical path represents the sequence of dependent activities that must be completed exactly on schedule to prevent any delay to the project's final delivery date.

A1 → A2 → B1 → B2.1 → B2.2 → B3 → B4 → C1

A1 → A2 → B1 → B2.1 → B2.2 → B3 → B4 → C4.1 → C4.2 → D2 → D3.1
→ D3.2 → D4 → D5 → E2.1 → E2.2 → E3 → F1 → F2.1 → F2.2

9 References

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Appendices

From our internships, we gathered data that are used in the report. The link for those data → <https://dms.uom.lk/s/TPLyNet4yxymjcr>